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Field Case Study Report MAKSA App Validation on Fresh Camel Semen (Dubai, UAE, 2025)

Date: 17 November 2025
Location: Dubai CRC (Camel Reproduction Center), UAE

Conducted by:		Verified by:
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Executive Summary / Conclusions

- **"Avare Biotech" App works reliably, Usability proven in the field**

App was installed, registered, and used in minutes by CRC staff — without our assistance. Five videos were processed within minutes, with stable feature detection despite sub-optimal optical conditions.

- **Motility correlation is strong**

Total motility aligns with CASA, confirming validated core tracking.

- **Concentration values differ**

Performance was influenced by CASA slide limitations, dilution ratio inconsistencies, and challenges in FOV volume calculation. These are known technical factors, and the team is already fixing them to improve concentration accuracy and overall system stability.

Objective

Validate the MAKSA mobile app on fresh camel semen and compare its readings to those of a benchmark commercial CASA system.

How the Test Was Performed

Sample	**Equipment**
• Species: Camelus dromedarius • Type: Fresh semen • Dilution used: **1:8** • CASA slide: Hamilton slide • Temperature of the hot stage: **38°C**	• CASA system: Hamilton • Microscope: basic laboratory model with a heated stage • Phone: Asif's personal IPhone 17 (Avare Biotech App powered by MAKSA technology installed) • Recording method: direct video capture through microscope eyepiece, no adapters or lighting enhancements — *deliberately field-realistic setup*

Procedure (transparent and simple)

1. Asif diluted the semen according to CRC's standard internal CASA protocol.
2. Semen was placed on the CASA slide and analysed first on the Hamilton CASA system.
3. Without changing the slide, the same CASA slide was moved to the **basic microscope**.
4. Asif recorded 5 short videos using MAKSA and uploaded them directly through the app.
5. MAKSA processed all videos in **under 5 minutes**.

This testing setup intentionally simulates real conditions — minimal equipment, natural workflow, and no specialized holders or adapters.

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Table 1MAKSA results comparison

	Parameters	MAKSA (Asif's video from CRC microscope)					Average	CASA (Hamilton)					Average	Difference, %
CASA slides	Cells number	55	44	62	43	59	52.6	277	293	259	289	274	278.4	429%
	Concentration M/ml	88	70.4	99.2	68.8	94.4	84.16	581.2	598	579.8	586.5	583.7	585.84	596%
	Total Motility %	15	17	20	14	16	16.4	11.9	11.8	12.7	12.9	13.4	12.54	-24%
	Progressive Motility %	12	14	17	12	13	13.6	5.4	6	6.5	7.2	7.2	6.46	-53%

Interpretation

Cell Count	Concentration
• CASA: 278 cells (avg) • MAKSA: ≈53 cells (avg) • 4–5× difference Reason: Different effective Field of View (FOV) — standard CASA optics capture a significantly larger area than a smartphone-through-eyepiece configuration.	• CASA: ~586 M/ml • MAKSA: ~84 M/ml • ≈6× difference This deviation was expected for two reasons: 1. Incorrect FOV volume calculation for camel semen (calibration is required). 2. Dilution factor (1:8) wasn't used

Total Motility	Progressive Motility
• CASA: 12.5% • MAKSA: 16% Good correlation. Slightly higher values on MAKSA observed.	• CASA: 6.4% • MAKSA: 13.6% • Difference exists but remains within known variability for fresh camel ejaculates when FOV geometry differs.

Key point:
Motility always improves automatically once concentration FOV correction is finalized — both metrics share the same underlying tracking model.

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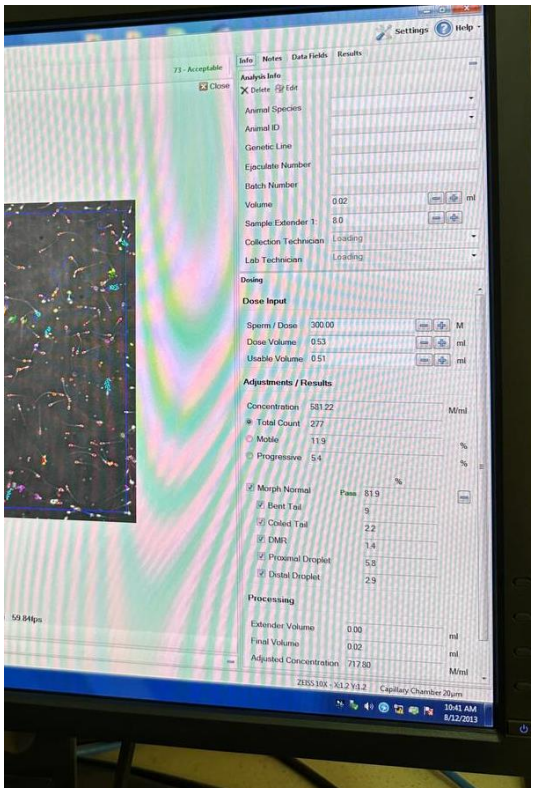
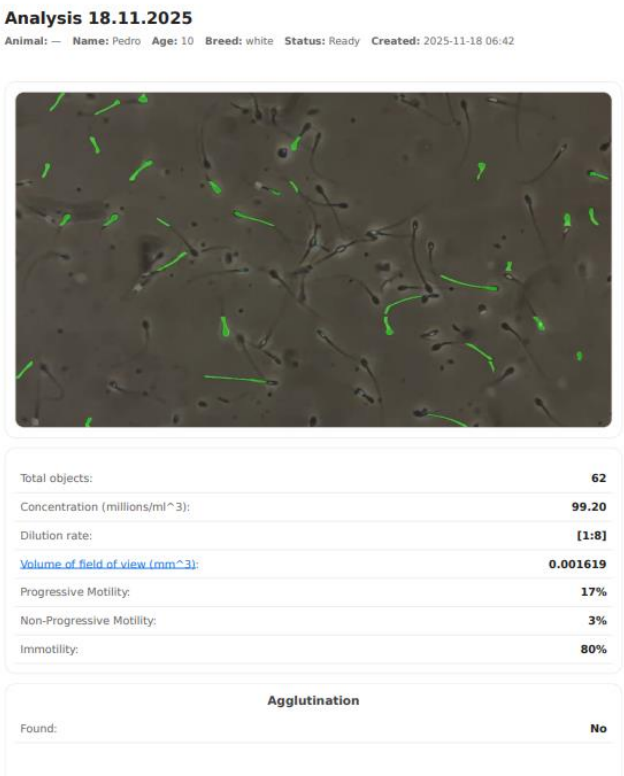
- The app requires species-specific volume calibration for correct concentration.
- A new FOV model will be created in the next 2-4 weeks.S
- Once standardization is complete, motility and concentration accuracy will rise together.
- Dilution workflow must be simplified in the app → we will automate dilution factor validation.

Appendix - Screenshots of CASA results and MAKSA reports

Camel 1: CASA readings	Camel 1: MAKSA values